Tasnim Chowdhury

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Education

Dartmouth College, Hanover, NH **June 2027**
Bachelor of Arts, Major in Computer Science, Minor in Human Centered Design **GPA 3.79/4.0**
Relevant Coursework: Algorithms & Data Structures, Scalable Systems, Object-Oriented Design, Web Development, Deep Learning, Mobile Development, Generative AI, Linear Algebra, Discrete Math

Relevant Experience

Salesforce | Digital Enterprise Technology **June 2026 – September 2026**
Intern, Sales Analytics & Agentic Infrastructure, San Francisco, CA

- Fixed undocumented double-count bugs in two production forecast models by defining table grain and usage rules; PR reviewed and merged into team dbt repo
- Built three Cortex Analyst semantic models (commit, sales targets, pipeline metrics) validated live against a certified executive dashboard
- Evaluated four architectures for LLM-driven conversational analytics; recommendation adopted as team's long-term approach
- Authored team's AI-agent context standard, adopted org-wide and cited as an onboarding enabler at a major platform go-live
- Won 1st place (300+ interns, 36 teams) at company hackathon for an Agentforce agent automating employee onboarding

OKC Thunder | Software Engineer Intern **incoming September 2026**

Lead Software Engineer | Production-Scale Rentals & Events Platform **June 2025 – September 2025**
Digital Applied Learning and Innovation (DALI) Lab, Hanover, NH

- Deployed platform on Render/Dokploy, later migrated to a containerized AWS stack (EC2, RDS/PostgreSQL, S3, Lambda) with Docker
- Refactored PostgreSQL/Prisma data layer, normalized hot paths, added composite indexes, cutting p95 latency 35%
- Implemented SSO auth with OAuth2, token rotation, and scoped permissions; built custom auth layer to remove Supabase
- Set up CI/CD via GitHub Actions with CloudWatch monitoring for zero-downtime releases

Lead Software Engineer | Full Stack Civic Resource Platform **Jan 2025 – March 2025**
Digital Applied Learning and Innovation (DALI) Lab, Hanover, NH

- Architected PostgreSQL backend supporting 10K+ community and environmental data points with optimized indexing
- Built interactive Mapbox interface in React + TypeScript for real-time air quality and community resource visualization
- Integrated weather/pollution APIs with caching, reducing load latency ~40%
- Built secure APIs with JWT/OAuth2 and rate limiting; implemented audit logging and access controls aligned with SOC2 readiness requirements

Projects & Leadership

Lead Full Stack Developer | Humanity First Gift of Health (GOH) **Feb 2025 – Present**

- Built offline-first EMR (React, Node.js, MongoDB) for disaster-relief clinics with limited connectivity
- Implemented HIPAA-aligned encryption, role-based access control, and multilingual patient/inventory modules

Dental X-Ray to 3D Model Pipeline **May 2026 – Present**

- Built SAM2-based auto-segmentation pipeline detecting all 28 teeth on real panoramic X-rays at >90% IoU
- Generated anatomically accurate 3D meshes lofted to patient-specific tooth contours; built full-mouth Three.js viewer with cavity markers and before/after treatment toggle
- Built DICOM ingestion pipeline with PHI de-identification per DICOM PS3.15; covered pipeline with 45 automated tests

Founder & Developer | Sook Sout — Arabic Dialect Learning App **2026 – Present**

- Built web (React, TypeScript, Vite, Tailwind) and mobile (Expo/React Native) apps for learning spoken Arabic dialects, serving 970 native-speaker recordings (Supabase) across a 153-phrase Jordanian corpus from 4 speakers
- Built Python audio pipeline (splitting, transcription, cataloging) with dialect-integrity rules so recordings are never mislabeled across dialects

Co-founder & Technical Developer Lead | Florev (Biotech Startup) **March 2024 – Present**

- Built non-invasive ischemic stroke detection system combining Arduino biosensors with a cloud AI pipeline (Python/FastAPI)
- Navigated IRB preparation and patent filing (patent filed) alongside the technical build

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL | **Web & App**: React, Next.js, Node.js, Express, Tailwind, HTML/CSS, Figma
Databases: PostgreSQL, MySQL, SQLite, MongoDB, Firebase | **Cloud & DevOps**: Snowflake, dbt, Tableau, AWS, GCP, Docker, GitHub Actions, CI/CD
AI/ML: Pandas, NumPy, scikit-learn, PyTorch, FastAPI, Hugging Face | **Certifications**: AWS Certified Cloud Practitioner, GCDLeader